Optimization In AI

Class 04: Stochastic Gradient Descent

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Today's Topics

01 Stochastic Gradients

02 Nonconvex Smooth Functions

03 Convex Nonsmooth Functions

04 Convex Smooth Functions

05 Strongly Convex Functions

06 Mini-Batches

Main Results

Setting	Step Size	Certificate
Nonconvex smooth	$\eta \asymp T^{-1/2}$ capped at $1/L$	Average $\mathbb{E}[\|\nabla f\|^2]$: $O(T^{-1/2})$
Convex nonsmooth	$\eta = R/(G\sqrt{T})$	Averaged-iterate gap: $RG/\sqrt{T}$
Convex smooth	$\eta = \min\{\frac{1}{2L}, \frac{R}{\sigma\sqrt{T}}\}$	Averaged-iterate gap: $O(\frac{LR^2}{T} + \frac{R\sigma}{\sqrt{T}})$
Strongly convex (also L-smooth)	Constant $\eta \le 1/L$ or $\eta_t = \frac{2}{\mu(t+\gamma)}$	Expected last gap: geometric + noise; $O(1/t)$

PART 01

Stochastic Gradients

Stochastic Gradient Descent

$$f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x), \qquad x \in \mathbb{R}^d.$$

Gradient Descent

$$x_{t+1} = x_t - \eta_t \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(x_t).$$

Stochastic Gradient Descent

$$i_t \sim \mathrm{Unif}\{1, \dots, n\}, \qquad \boxed{x_{t+1} = x_t - \eta_t \nabla f_{i_t}(x_t).}$$

Sampling With And Without Replacement

With Replacement: Independent Draws

$i_t \overset{\text{iid}}{\sim} \text{Unif}\{1, \ldots, n\}$.

$$n = 8: \qquad 2,\ 5,\ 2,\ 1,\ 5,\ 8,\ 3,\ 2, \ldots$$

Without Replacement: A Random Permutation

$$\pi = (3,\ 1,\ 4,\ 2,\ 8,\ 5,\ 7,\ 6).$$

Mini-Batches And Data Passes

A Mini-Batch: B Examples, One Update

$$g_t = \frac{1}{B}\sum_{j=1}^{B} \nabla f_{i_{t,j}}(x_t), \qquad x_{t+1} = x_t - \eta_t g_t.$$



$n = 8,\ B = 2.$

Data Traversal	Examples Processed	Updates ($B \mid n$)
One pass (one epoch)	Each example once: n	n/B
Multiple passes: K epochs	Each example K times: Kn	Kn/B

The Stochastic Gradient Oracle

$\mathcal{F}_t$: history before the fresh sample; x_t is $\mathcal{F}_t$-measurable.

$$\mathbb{E}_t[\,\cdot\,] := \mathbb{E}[\,\cdot \mid \mathcal{F}_t], \qquad x_{t+1} = x_t - \eta_t g_t.$$

Assumption (Unbiasedness And Bounded Variance)

For every $t \geq 0$, almost surely,

$$\mathbb{E}_t[g_t] = \nabla f(x_t), \qquad \mathbb{E}_t[\|g_t - \nabla f(x_t)\|^2] \leq \sigma^2.$$

$$g_t = \nabla f(x_t) + \varepsilon_t, \qquad \mathbb{E}_t[\varepsilon_t] = 0.$$

Fresh uniform sampling: $\mathbb{E}_t[\nabla f_{i_t}(x_t)] = \dfrac{1}{n}\sum_i \nabla f_i(x_t) = \nabla f(x_t).$

Example: Stochastic Linear Regression

$$f(x) = \frac{1}{2n}\sum_{i=1}^{n}(a_i^\top x - b_i)^2, \qquad g_t = (a_{i_t}^\top x_t - b_{i_t})a_{i_t}.$$

An Unbiased Direction Can Point Uphill

$a_1 = a_2 = 1,\ b_1 = 1,\ b_2 = -1$: $\quad f(x) = \frac{1}{2}x^2 + \frac{1}{2}$.

$$x_t = \tfrac{1}{4}, \qquad g_t = \begin{cases} -3/4, & i_t = 1, \\ 5/4, & i_t = 2, \end{cases} \qquad \mathbb{E}_t[g_t] = \tfrac{1}{4} = \nabla f(x_t).$$

$$i_t = 1, \quad \eta = 0.1 \quad \Longrightarrow \quad x_{t+1} = 0.325, \qquad f(x_{t+1}) > f(x_t).$$

PART 02

Nonconvex Smooth Functions

The Stochastic Descent Lemma

Lemma (Stochastic Descent Lemma)

Let $f : \mathbb{R}^d \to \mathbb{R}$ be L-smooth, $L > 0$, and let the oracle be conditionally unbiased with variance at most σ^2. For a deterministic $\eta_t > 0$,

$$\mathbb{E}_t[f(x_{t+1})] \leq f(x_t) - \eta_t \left(1 - \frac{L\eta_t}{2}\right) \|\nabla f(x_t)\|^2 + \frac{L\eta_t^2\sigma^2}{2}.$$

For $0 < \eta_t \leq 1/L$,

$$\boxed{\mathbb{E}_t[f(x_{t+1})] \leq f(x_t) - \frac{\eta_t}{2}\|\nabla f(x_t)\|^2 + \frac{L\eta_t^2\sigma^2}{2}.}$$

Proof: Use The Smooth Upper Model

Smooth Upper Model: $x_{t+1} = x_t - \eta_t g_t$

$$f(x_{t+1}) \leq f(x_t) - \eta_t \langle \nabla f(x_t), g_t \rangle + \frac{L\eta_t^2}{2} \|g_t\|^2.$$

Unbiasedness Resolves The Linear Term

$$\mathbb{E}_t[\langle \nabla f(x_t), g_t \rangle] = \langle \nabla f(x_t), \mathbb{E}_t[g_t] \rangle = \|\nabla f(x_t)\|^2.$$

$$\mathbb{E}_t[f(x_{t+1})] \leq f(x_t) - \eta_t \|\nabla f(x_t)\|^2 + \frac{L\eta_t^2}{2} \mathbb{E}_t[\|g_t\|^2].$$

Proof: Separate Signal And Noise

Variance Identity: $g_t = \nabla f(x_t) + \varepsilon_t$, $\mathbb{E}_t[\varepsilon_t] = 0$

$$\begin{aligned}\mathbb{E}_t[\|g_t\|^2] &= \|\nabla f(x_t)\|^2 + 2\left\langle \nabla f(x_t), \underbrace{\mathbb{E}_t[\varepsilon_t]}_{0} \right\rangle + \mathbb{E}_t[\|\varepsilon_t\|^2] \\ &\leq \|\nabla f(x_t)\|^2 + \sigma^2.\end{aligned}$$

Combine With The Expected Upper Model

$$\mathbb{E}_t[f(x_{t+1})] \leq f(x_t) - \eta_t \left(1 - \frac{L\eta_t}{2}\right) \|\nabla f(x_t)\|^2 + \frac{L\eta_t^2\sigma^2}{2}.$$

$$\eta_t \leq 1/L \quad \Longrightarrow \quad 1 - \frac{L\eta_t}{2} \geq \frac{1}{2}.$$

Nonconvex SGD: Stationarity Rate

Theorem (Nonconvex SGD: Stationarity Bound)

Let f be L-smooth, $f^\star := \inf f > -\infty$, and $\Delta_0 = f(x_0) - f^\star$. Assume $\mathbb{E}_t[g_t] = \nabla f(x_t)$ and $\mathbb{E}_t[\|g_t - \nabla f(x_t)\|^2] \le \sigma^2$. For constant $0 < \eta \le 1/L$ and every integer $T \ge 1$,

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(x_t)\|^2] \le \frac{2\Delta_0}{\eta T} + L\eta\sigma^2.$$

Balanced Step: $\Delta_0 > 0$, $\sigma > 0$

$$\eta = \sqrt{\frac{2\Delta_0}{L\sigma^2 T}} \le \frac{1}{L} \quad \Longrightarrow \quad \frac{1}{T}\sum_{t<T} \mathbb{E}[\|\nabla f(x_t)\|^2] \le 2\sqrt{\frac{2L\Delta_0\sigma^2}{T}} = O(T^{-1/2}).$$

Proof: Telescope The Noisy Decrease

Take Full Expectation, Then Sum

$$\frac{\eta}{2}\mathbb{E}[\|\nabla f(x_t)\|^2] \le \mathbb{E}[f(x_t) - f(x_{t+1})] + \frac{L\eta^2\sigma^2}{2}.$$

$$\frac{\eta}{2}\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla f(x_t)\|^2] \le f(x_0) - \mathbb{E}[f(x_T)] + \frac{TL\eta^2\sigma^2}{2}$$

$$\le \Delta_0 + \frac{TL\eta^2\sigma^2}{2}.$$

Divide By $\eta T/2$

$$\boxed{\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla f(x_t)\|^2] \le \frac{2\Delta_0}{\eta T} + L\eta\sigma^2.}$$

Potential View: Nonconvex Stationarity

$\Delta_t = f(x_t) - f^\star \geq 0, \quad \text{fixed } T \geq 1, \quad 0 < \eta \leq 1/L.$

Definition (Nonconvex SGD Potential)

$$\Phi_t = \Delta_t + \frac{\eta}{2} \sum_{s=0}^{t-1} \|\nabla f(x_s)\|^2 + \frac{(T-t)L\eta^2\sigma^2}{2}, \quad 0 \leq t \leq T.$$

$$\Phi_0 = \Delta_0 + \frac{TL\eta^2\sigma^2}{2}, \qquad \Phi_T \geq \frac{\eta}{2} \sum_{t=0}^{T-1} \|\nabla f(x_t)\|^2.$$

Proof: The Nonconvex Potential

Substitute The Stochastic Descent Bound

$$\begin{aligned}\mathbb{E}_t[\Phi_{t+1}] - \Phi_t &= \mathbb{E}_t[\Delta_{t+1}] - \Delta_t + \frac{\eta}{2}\|\nabla f(x_t)\|^2 - \frac{L\eta^2\sigma^2}{2} \\ &\leq 0.\end{aligned}$$

Compare Endpoints And Divide By $\eta T/2$

$$\frac{\eta}{2}\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla f(x_t)\|^2] \leq \mathbb{E}[\Phi_T] \leq \Phi_0 = \Delta_0 + \frac{TL\eta^2\sigma^2}{2}.$$

$$\boxed{\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla f(x_t)\|^2] \leq \frac{2\Delta_0}{\eta T} + L\eta\sigma^2.}$$

Balance Progress And Noise

$$\frac{2\Delta_0}{\eta T} + L\eta\sigma^2$$

For $\Delta_0 > 0$ And $\sigma > 0$

$$\eta = \min\left\{\frac{1}{L}, \sqrt{\frac{2\Delta_0}{L\sigma^2 T}}\right\}.$$

$\sqrt{2\Delta_0/(L\sigma^2 T)} \le 1/L$:

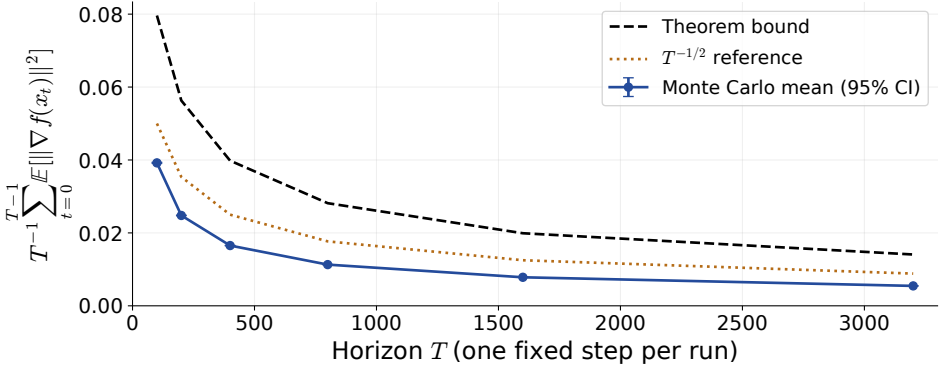
$$\boxed{\frac{1}{T}\sum_{t<T}\mathbb{E}[\|\nabla f(x_t)\|^2] \le 2\sqrt{\frac{2L\Delta_0\sigma^2}{T}} = O(T^{-1/2}).}$$

$$\sigma = 0, \quad \eta = 1/L \quad \Longrightarrow \quad \tfrac{1}{T}\textstyle\sum_{t<T}\mathbb{E}[\|\nabla f(x_t)\|^2] \le \frac{2L\Delta_0}{T}.$$

Simulation: Nonconvex Stationarity

$$f(x) = 1 - \cos x_1 + \tfrac{1}{50}(1 - \cos x_2), \quad x_0 = (2.8, 2.8)^\top.$$

$L = 1$, $\varepsilon_t \sim \mathcal{N}(0, 0.02 I_2)$; $\eta = \min\{1, \sqrt{2\Delta_0/(0.04\,T)}\}$; 2,000 runs.



PART 03

Convex Nonsmooth Functions

Stochastic Subgradient Method: Convergence

Theorem (Stochastic Subgradient Method: Convex Convergence)

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex with minimizer $x^\star$ and $\|x_0 - x^\star\| \le R$. For every $t \ge 0$, almost surely, assume

$$h_t := \mathbb{E}_t[g_t] \in \partial f(x_t), \qquad \mathbb{E}_t[\|g_t\|^2] \le G^2.$$

For $x_{t+1} = x_t - \eta g_t$, constant $\eta > 0$, every integer $T \ge 1$, and $\bar{x}_T = T^{-1} \sum_{t=0}^{T-1} x_t$,

$$\boxed{\mathbb{E}[f(\bar{x}_T)] - f^\star \le \frac{R^2}{2\eta T} + \frac{\eta G^2}{2}.}$$

$$R, G > 0, \quad \eta = \frac{R}{G\sqrt{T}} \quad \Longrightarrow \quad \mathbb{E}[f(\bar{x}_T)] - f^\star \le \frac{RG}{\sqrt{T}}.$$

Proof: The Stochastic Subgradient Bound

$D_t := \|x_t - x^\star\|, \quad \Delta_t := f(x_t) - f^\star, \quad h_t := \mathbb{E}_t[g_t] \in \partial f(x_t).$

Expand The Distance; Apply The Subgradient Inequality

$$\begin{aligned}\mathbb{E}_t[D_{t+1}^2] &= D_t^2 - 2\eta \langle h_t, x_t - x^\star \rangle + \eta^2 \mathbb{E}_t[\|g_t\|^2] \\ &\leq D_t^2 - 2\eta\Delta_t + \eta^2 G^2.\end{aligned}$$

Take Expectation, Telescope, Then Apply Jensen

$$2\eta \sum_{t<T} \mathbb{E}[\Delta_t] \leq D_0^2 - \mathbb{E}[D_T^2] + T\eta^2 G^2 \leq R^2 + T\eta^2 G^2.$$

$$\mathbb{E}[f(\bar{x}_T)] - f^\star \leq \frac{1}{T} \sum_{t<T} \mathbb{E}[\Delta_t] \leq \boxed{\frac{R^2}{2\eta T} + \frac{\eta G^2}{2}}.$$

Potential View: Convex Nonsmooth SGD

$D_t = \|x_t - x^\star\|, \quad \Delta_t = f(x_t) - f^\star, \quad \text{fixed } T \geq 1,\ \eta > 0.$

Subgradient Distance Recurrence

$$\mathbb{E}_t[D_{t+1}^2] \leq D_t^2 - 2\eta\Delta_t + \eta^2 G^2.$$

Definition (Stochastic Subgradient Potential)

$$\Phi_t = D_t^2 + 2\eta \sum_{s=0}^{t-1} \Delta_s + (T-t)\eta^2 G^2, \qquad 0 \leq t \leq T.$$

$$\Phi_0 = D_0^2 + T\eta^2 G^2, \qquad \Phi_T = D_T^2 + 2\eta \sum_{t=0}^{T-1} \Delta_t.$$

Proof: The Stochastic Subgradient Potential

Cancel The Gap And Second-Moment Terms

$$\begin{aligned}\mathbb{E}_t[\Phi_{t+1}] - \Phi_t &= \mathbb{E}_t[D_{t+1}^2] - D_t^2 + 2\eta\Delta_t - \eta^2 G^2 \\ &\le (-2\eta\Delta_t + \eta^2 G^2) + 2\eta\Delta_t - \eta^2 G^2 = 0.\end{aligned}$$

Endpoints And Jensen: $\bar{x}_T = T^{-1}\sum_{t<T} x_t$

$$2\eta \sum_{t=0}^{T-1} \mathbb{E}[\Delta_t] \le \mathbb{E}[\Phi_T] \le \Phi_0 \le R^2 + T\eta^2 G^2.$$

$$\boxed{\mathbb{E}[f(\bar{x}_T)] - f^\star \le \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\Delta_t] \le \frac{R^2}{2\eta T} + \frac{\eta G^2}{2}.}$$

PART 04

Convex Smooth Functions

Convex Smooth SGD: Convergence Rate

Theorem (Convex Smooth SGD: Averaged-Iterate Convergence)

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and L-smooth, $L > 0$, with minimizer $x^\star$ and $\|x_0 - x^\star\| \le R$. Assume, for every $t \ge 0$, almost surely,

$$\mathbb{E}_t[g_t] = \nabla f(x_t), \qquad \mathbb{E}_t[\|g_t - \nabla f(x_t)\|^2] \le \sigma^2.$$

For $x_{t+1} = x_t - \eta g_t$, constant $0 < \eta \le 1/(2L)$, every integer $T \ge 1$, and $\bar{x}_T = T^{-1} \sum_{t=0}^{T-1} x_t$,

$$\boxed{\mathbb{E}[f(\bar{x}_T)] - f^\star \le \frac{R^2}{\eta T} + \eta \sigma^2.}$$

Proof: Control The Gradient By The Gap

Apply The Smooth Upper Model To An Exact Gradient Step

For any $x \in \mathbb{R}^d$, set $y = x - \nabla f(x)/L$:

$$f^\star \le f(y) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2.$$

$$\boxed{\|\nabla f(x)\|^2 \le 2L\big(f(x) - f^\star\big).}$$

Bound The Second Moment: $\Delta_t := f(x_t) - f^\star$

$$\mathbb{E}_t[\|g_t\|^2] = \|\nabla f(x_t)\|^2 + \mathbb{E}_t[\|g_t - \nabla f(x_t)\|^2] \le 2L\Delta_t + \sigma^2.$$

Proof: Telescope The Convex Smooth Bound

$D_t := \|x_t - x^\star\|, \quad \Delta_t := f(x_t) - f^\star.$

Convexity And The Second-Moment Bound

$$\begin{aligned}\mathbb{E}_t[D_{t+1}^2] &\le D_t^2 - 2\eta(1 - L\eta)\Delta_t + \eta^2\sigma^2 \\ &\le D_t^2 - \eta\Delta_t + \eta^2\sigma^2 \qquad (\eta \le 1/(2L)).\end{aligned}$$

Take Expectation, Sum, And Average

$$\eta \sum_{t<T} \mathbb{E}[\Delta_t] \le D_0^2 - \mathbb{E}[D_T^2] + T\eta^2\sigma^2 \le R^2 + T\eta^2\sigma^2.$$

$$\boxed{\mathbb{E}[f(\bar{x}_T)] - f^\star \le \frac{1}{T}\sum_{t<T} \mathbb{E}[\Delta_t] \le \frac{R^2}{\eta T} + \eta\sigma^2.}$$

Potential View: Convex Smooth SGD

$D_t = \|x_t - x^\star\|, \quad \Delta_t = f(x_t) - f^\star, \quad \text{fixed } T \geq 1.$

Smoothness Changes The Distance Recurrence

$$\mathbb{E}_t[\|g_t\|^2] \leq 2L\Delta_t + \sigma^2, \qquad 0 < \eta \leq 1/(2L).$$
$$\mathbb{E}_t[D_{t+1}^2] \leq D_t^2 - \eta\Delta_t + \eta^2\sigma^2.$$

Definition (Convex Smooth SGD Potential)

$$\Phi_t = D_t^2 + \eta \sum_{s=0}^{t-1} \Delta_s + (T-t)\eta^2\sigma^2, \qquad 0 \leq t \leq T.$$

Proof: The Convex Smooth Potential

Conditional Change And Initial Value

$$\mathbb{E}_t[\Phi_{t+1}] - \Phi_t = \mathbb{E}_t[D_{t+1}^2] - D_t^2 + \eta\Delta_t - \eta^2\sigma^2 \le 0.$$
$$\Phi_0 = D_0^2 + T\eta^2\sigma^2 \le R^2 + T\eta^2\sigma^2.$$

Terminal Value And Jensen

$$\eta \sum_{t=0}^{T-1} \mathbb{E}[\Delta_t] \le \mathbb{E}\left[D_T^2 + \eta \sum_{t=0}^{T-1} \Delta_t\right] = \mathbb{E}[\Phi_T] \le \Phi_0.$$

$$\boxed{\mathbb{E}[f(\bar{x}_T)] - f^\star \le \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\Delta_t] \le \frac{R^2}{\eta T} + \eta\sigma^2.}$$

Convex Smooth SGD: Step Size And Noise

Balance The Bound, Subject To The Smoothness Cap

For $R, \sigma > 0$, choose $\eta = \min\{1/(2L), R/(\sigma\sqrt{T})\}$.

$$\mathbb{E}[f(\bar{x}_T)] - f^\star \le \begin{cases} 2R\sigma/\sqrt{T}, & R/(\sigma\sqrt{T}) \le 1/(2L), \\ 2LR^2/T + \sigma^2/(2L), & R/(\sigma\sqrt{T}) > 1/(2L). \end{cases}$$

In the second case, $\sigma^2/(2L) < R\sigma/\sqrt{T}$. Hence, in both cases,

$$\boxed{\mathbb{E}[f(\bar{x}_T)] - f^\star \le \frac{2LR^2}{T} + \frac{2R\sigma}{\sqrt{T}}.}$$

Zero Noise

$$\sigma = 0, \quad \eta = 1/(2L) \quad \Longrightarrow \quad \mathbb{E}[f(\bar{x}_T)] - f^\star \le \frac{2LR^2}{T}.$$

PART 05

Strongly Convex Functions

Constant Steps: Transient And Floor

$\Delta_t := f(x_t) - f^\star$, where $x^\star$ is the unique minimizer.

Theorem (Strongly Convex SGD: Constant Step Size)

Let f be L-smooth and μ-strongly convex, $0 < \mu \le L$. Assume $\mathbb{E}_t[g_t] = \nabla f(x_t)$ and $\mathbb{E}_t[\|g_t - \nabla f(x_t)\|^2] \le \sigma^2$. For constant $0 < \eta \le 1/L$, $q = 1 - \mu\eta \in [0, 1)$, and every integer $T \ge 1$,

$$\mathbb{E}[\Delta_T] \le q^T \Delta_0 + \frac{L\eta\sigma^2}{2\mu}(1 - q^T) \le q^T \Delta_0 + \frac{L\eta\sigma^2}{2\mu}.$$

Proof: The Same Contraction Plus Noise

Strong Convexity: Gap–Gradient Inequality

$$\|\nabla f(x_t)\|^2 \geq 2\mu\Delta_t, \qquad \Delta_t = f(x_t) - f^\star.$$

Substitute Into The Stochastic Descent Lemma

$$\begin{aligned}\mathbb{E}_t[\Delta_{t+1}] &\leq \Delta_t - \frac{\eta}{2}\|\nabla f(x_t)\|^2 + \frac{L\eta^2\sigma^2}{2} \\ &\leq (1-\mu\eta)\Delta_t + \frac{L\eta^2\sigma^2}{2}.\end{aligned}$$

Proof: Unroll The Recurrence

$$a_t := \mathbb{E}[\Delta_t], \qquad q := 1 - \mu\eta, \qquad c := L\eta^2\sigma^2/2.$$

Unroll The Recurrence

$$a_{t+1} \le q a_t + c \quad \Longrightarrow \quad a_T \le q^T a_0 + c \sum_{k=0}^{T-1} q^k.$$

Sum The Geometric Series: $1 - q = \mu\eta > 0$

$$\sum_{k=0}^{T-1} q^k = \frac{1 - q^T}{1 - q} \le \frac{1}{\mu\eta}.$$

$$\boxed{a_T \le (1 - \mu\eta)^T \Delta_0 + \frac{L\eta\sigma^2}{2\mu}.}$$

Potential View: Constant-Step SGD

$\Delta_t = f(x_t) - f^\star, \quad q = 1 - \mu\eta \in [0,1), \quad c = L\eta^2\sigma^2/2.$

$T \geq 1$ fixed, $\qquad q^0 := 1$ for all $q \in [0,1), \qquad$ empty sum $= 0$.

Definition (Constant-Step SGD Terminal Potential)

$$\Phi_t = q^{T-t}\Delta_t + c \sum_{k=0}^{T-t-1} q^k, \qquad 0 \leq t \leq T.$$

$$\Phi_T = \Delta_T, \qquad \Phi_0 = q^T\Delta_0 + c\sum_{k=0}^{T-1} q^k.$$

Proof: Geometric Weights And The Noise Floor

Use $\mathbb{E}_t[\Delta_{t+1}] \le q\Delta_t + c$: $0 \le t < T$

$$\mathbb{E}_t[\Phi_{t+1}] - \Phi_t = q^{T-t-1}\big(\mathbb{E}_t[\Delta_{t+1}] - q\Delta_t - c\big) \le 0.$$

Evaluate The Initial Budget: $1 - q = \mu\eta > 0$

$$\mathbb{E}[\Delta_T] \le \Phi_0 = q^T\Delta_0 + \frac{c(1-q^T)}{1-q}.$$

$$\boxed{\mathbb{E}[\Delta_T] \le (1-\mu\eta)^T\Delta_0 + \frac{L\eta\sigma^2}{2\mu}\big(1-(1-\mu\eta)^T\big).}$$

$$q = 0: \qquad \mathbb{E}[\Delta_T] \le c \qquad (T \ge 1).$$

Decreasing Steps Recover $1/T$

Theorem (Strongly Convex SGD: Decreasing Step Sizes)

Let f be L-smooth and μ-strongly convex, $0 < \mu \leq L$, and assume the unbiased oracle has conditional variance at most σ^2. Choose

$$\eta_t = \frac{2}{\mu(t+\gamma)}, \qquad \gamma \geq \frac{2L}{\mu}, \qquad A \geq \max\left\{\gamma\Delta_0, \frac{2L\sigma^2}{\mu^2}\right\}.$$

Then, for every integer $t \geq 0$,

$$\boxed{\mathbb{E}[f(x_t) - f^\star] \leq \frac{A}{t+\gamma}.}$$

Proof: Match The Recurrence

$$s = t + \gamma \geq 2, \qquad C = \frac{2L\sigma^2}{\mu^2}, \qquad a_t = \mathbb{E}[\Delta_t].$$

Base: $A \geq \gamma \Delta_0 \implies a_0 \leq A/\gamma$.

Induction: Assume $a_t \leq A/s$

$$\begin{aligned}
a_{t+1} &\leq \left(1 - \frac{2}{s}\right) a_t + \frac{C}{s^2} \\
&\leq \frac{A}{s} - \frac{2A - C}{s^2} && \text{since } s \geq 2,\ a_t \leq A/s \\
&\leq \frac{A}{s} - \frac{A}{s^2} && \text{since } A \geq C \\
&= \frac{A(s-1)}{s^2} \leq \frac{A}{s+1} && \text{since } (s-1)(s+1) \leq s^2.
\end{aligned}$$

Potential View: Scale By The Target Rate

$\eta_t = 2/[\mu(t+\gamma)], \quad \gamma \geq 2L/\mu \geq 2, \quad C = 2L\sigma^2/\mu^2.$

$$\mathbb{E}_t[\Delta_{t+1}] \leq \left(1 - \frac{2}{t+\gamma}\right)\Delta_t + \frac{C}{(t+\gamma)^2}.$$

Definition (Rate-Scaled Expected Gap)

$$\Psi_t = (t+\gamma)\mathbb{E}[\Delta_t].$$

$$\mathbb{E}[\Delta_t] \leq \frac{A}{t+\gamma} \quad \Longleftrightarrow \quad \Psi_t \leq A.$$

Invariant Bound

$$\Psi_0 = \gamma\Delta_0 \leq A, \qquad \Psi_t \leq A \implies \Psi_{t+1} \leq A.$$

Proof: The Scaled Potential Stays Bounded

$s = t + \gamma \geq 2, \quad A \geq \max\{\gamma\Delta_0, C\}.$

Rescale The Expected Recurrence

$$\Psi_{t+1} \leq \frac{(s+1)(s-2)}{s^2}\Psi_t + \frac{C(s+1)}{s^2}.$$

Preserve $\Psi_t \leq A$: The Multiplier Is Nonnegative

$$\begin{aligned}\Psi_{t+1} &\leq \frac{(s+1)(s-2)}{s^2}A + \frac{C(s+1)}{s^2} \\ &\leq A\frac{(s+1)(s-1)}{s^2} = A\left(1 - \frac{1}{s^2}\right) \leq A.\end{aligned}$$

$$\Psi_0 = \gamma\Delta_0 \leq A \quad \Longrightarrow \quad \boxed{\mathbb{E}[\Delta_t] \leq \frac{A}{t+\gamma} \quad (t \geq 0).}$$

Potential View: Error Relative To The Target

$s = t + \gamma \geq 2, \quad C = 2L\sigma^2/\mu^2, \quad A \geq \max\{\gamma\Delta_0, C\}.$

Definition (Centered Gap Relative To The Rate Bound)

$$Z_t = \Delta_t - \frac{A}{t+\gamma}.$$

Subtract The Next Target Value

$$\begin{aligned}\mathbb{E}_t[Z_{t+1}] &\leq \left(1 - \frac{2}{s}\right)\left(Z_t + \frac{A}{s}\right) + \frac{C}{s^2} - \frac{A}{s+1} \\ &= \left(1 - \frac{2}{s}\right) Z_t - \frac{A - C}{s^2} - \frac{A}{s^2(s+1)} \leq \left(1 - \frac{2}{s}\right) Z_t.\end{aligned}$$

$$\mathbb{E}[Z_0] \leq 0 \implies \mathbb{E}[Z_t] \leq 0 \implies \mathbb{E}[\Delta_t] \leq A/(t+\gamma).$$

Potential View: Decreasing-Step SGD

$H_T = (T+\gamma-2)(T+\gamma-1), \quad C = 2L\sigma^2/\mu^2.$

Definition (Decreasing-Step SGD Terminal Potential)

For fixed $T \geq 1$, $\gamma \geq 2L/\mu \geq 2$, and $0 \leq t \leq T$,

$$\Phi_t = \frac{(t+\gamma-2)(t+\gamma-1)\Delta_t + C\sum_{k=t}^{T-1} \frac{k+\gamma-1}{k+\gamma}}{H_T}.$$

Substitute The Decreasing-Step Recurrence: $s = t+\gamma$, $t < T$

$$\mathbb{E}_t[\Phi_{t+1}] - \Phi_t = \frac{(s-1)s}{H_T}\left(\mathbb{E}_t[\Delta_{t+1}] - \frac{s-2}{s}\Delta_t - \frac{C}{s^2}\right) \leq 0.$$

$$\Phi_T = \Delta_T \qquad \Longrightarrow \qquad \mathbb{E}[\Delta_T] \leq \Phi_0.$$

Decreasing Steps: An Explicit Endpoint Bound

$C = 2L\sigma^2/\mu^2, \quad H_T = (T+\gamma-2)(T+\gamma-1) > 0.$

Corollary (Decreasing-Step SGD: Finite-Time Gap Bound)

Under the decreasing-step theorem's assumptions, for every integer $T \geq 1$,

$$\mathbb{E}[\Delta_T] \leq \frac{(\gamma-2)(\gamma-1)\Delta_0 + C\sum_{k=0}^{T-1} \frac{k+\gamma-1}{k+\gamma}}{H_T}.$$

Bound The Remaining Noise Sum

$$0 \leq \frac{k+\gamma-1}{k+\gamma} \leq 1 \quad \Longrightarrow \quad \mathbb{E}[\Delta_T] \leq \underbrace{\frac{(\gamma-2)(\gamma-1)\Delta_0}{H_T}}_{O(T^{-2})} + \underbrace{\frac{CT}{H_T}}_{O(T^{-1})}.$$

$$\gamma = 2: \qquad \mathbb{E}[\Delta_T] \leq C/(T+1).$$

Example: A Noisy Quadratic

Example (Same Objective, Different Noise Controls)

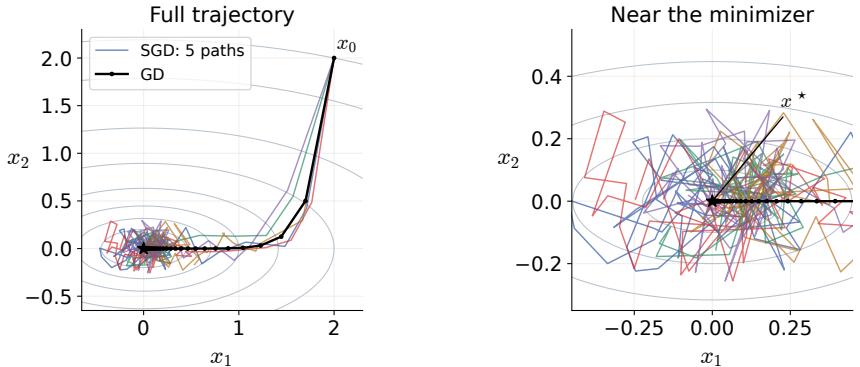
$$f(x) = \tfrac{1}{2}(x_1^2 + 5x_2^2), \qquad Q = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}, \qquad x_0 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}.$$

$$x^\star = 0, \qquad f^\star = 0, \qquad \mu = 1, \quad L = 5, \quad \Delta_0 = 12.$$

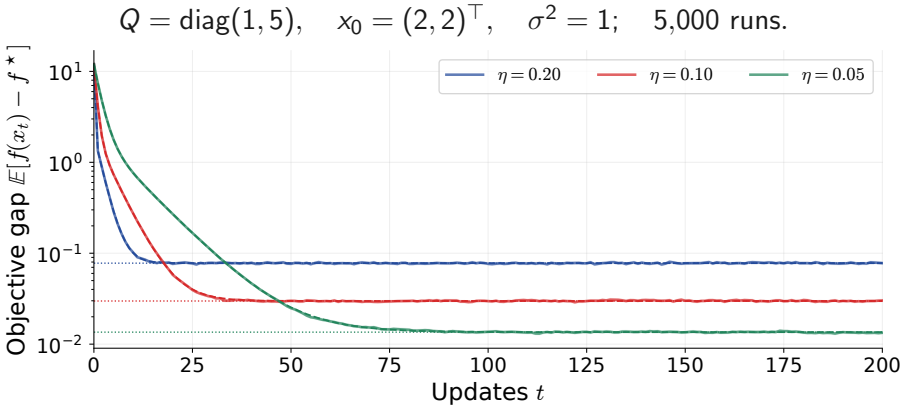
$$g_t = Qx_t + \varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \tfrac{1}{2} I_2), \qquad \mathbb{E}_t[\|\varepsilon_t\|^2] = \operatorname{tr}(\tfrac{1}{2} I_2) = 1.$$

Simulation: GD Versus SGD

$$Q = \mathrm{diag}(1, 5), \quad x_0 = (2, 2)^\top, \quad \eta = 0.15, \quad \sigma^2 = 1.$$

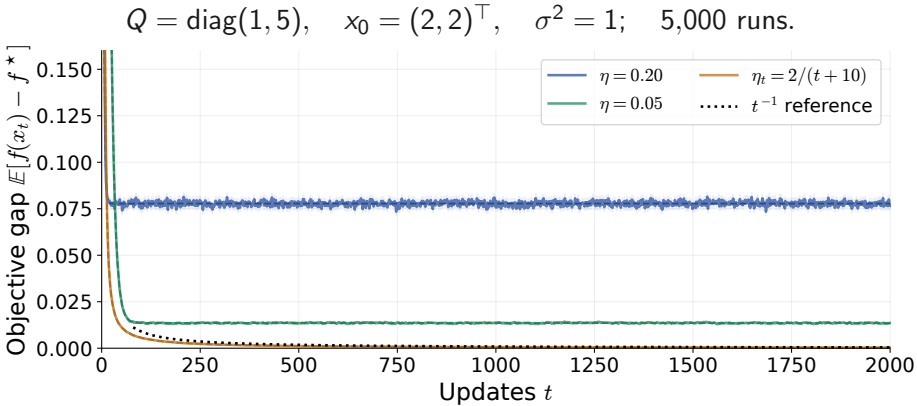


Simulation: Constant-Step Noise Floors



Solid: Monte Carlo; dashed: exact expectation; dotted: exact floor; shading: 95% CI.

Simulation: Decay Removes The Floor



Solid: Monte Carlo mean; dashed: exact expectation; shading: pointwise 95% CI.

PART 06

Mini-Batches

Mini-Batch SGD

Definition (Mini-Batch Stochastic Gradient)

Given $\mathcal{F}_t$, draw $B \geq 1$ conditionally independent oracle samples at x_t:

$$g_t^{(B)} = \frac{1}{B} \sum_{j=1}^{B} g(x_t; \zeta_{t,j}), \qquad x_{t+1} = x_t - \eta_t g_t^{(B)}.$$

Lemma (Mini-Batch Gradient: Unbiasedness And Variance)

If each sample is conditionally unbiased with variance at most σ^2, then

$$\mathbb{E}_t[g_t^{(B)}] = \nabla f(x_t), \qquad \mathbb{E}_t[\|g_t^{(B)} - \nabla f(x_t)\|^2] \leq \frac{\sigma^2}{B}.$$

Proof: Variance Shrinks As $1/B$

Let $\varepsilon_j = g(x_t; \zeta_{t,j}) - \nabla f(x_t)$; $\mathbb{E}_t[\varepsilon_j] = 0$ and $\mathbb{E}_t[\|\varepsilon_j\|^2] \le \sigma^2$.
$\mathbb{E}_t[g_t^{(B)}] = B^{-1} \sum_j \mathbb{E}_t[g(x_t; \zeta_{t,j})] = \nabla f(x_t)$.

Conditional Independence Kills The Cross Terms

$$\mathbb{E}_t\left[\left\|\frac{1}{B}\sum_j \varepsilon_j\right\|^2\right] = \frac{1}{B^2}\sum_j \mathbb{E}_t[\|\varepsilon_j\|^2] + \frac{1}{B^2}\sum_{i\neq j} \underbrace{\mathbb{E}_t[\langle \varepsilon_i, \varepsilon_j\rangle]}_{0}$$

$$\le \frac{B\sigma^2}{B^2} = \boxed{\frac{\sigma^2}{B}}.$$

Boundary: $\varepsilon_1 = \cdots = \varepsilon_B \implies B^{-1}\sum_j \varepsilon_j = \varepsilon_1$.

Mini-Batch Convergence Bounds

Conditionally Independent Samples

Smooth Nonconvex: $0 < \eta \leq 1/L$

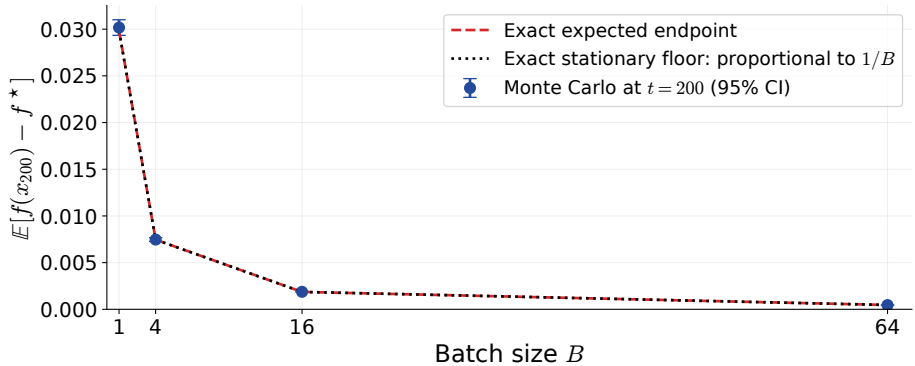
$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla f(x_t)\|^2] \leq \frac{2\Delta_0}{\eta T} + \frac{L\eta\sigma^2}{B}.$$

Smooth Strongly Convex: Same Step Restriction

$$\mathbb{E}[\Delta_T] \leq (1-\mu\eta)^T\Delta_0 + \frac{L\eta\sigma^2}{2\mu B}.$$

Simulation: Batch Size Reduces Noise

$Q = \mathrm{diag}(1, 5), \quad \eta = 0.1, \quad T = 200, \quad B = 1, 4, 16, 64; \quad 5{,}000$ runs.



Batch Size And Gradient Work

$$N = BT, \qquad \frac{1}{T}\sum_{t<T}\mathbb{E}[\|\nabla f(x_t)\|^2] \le \frac{2\Delta_0 B}{\eta N} + \frac{L\eta\sigma^2}{B}.$$

Balance At Fixed N: $\Delta_0, \sigma > 0$

$$\eta_* = B\sqrt{\frac{2\Delta_0}{L\sigma^2 N}} \le \frac{1}{L} \quad \Longrightarrow \quad \text{bound} = \boxed{2\sqrt{\frac{2L\Delta_0\sigma^2}{N}}}.$$

Beyond The Step Cap

$$\eta = 1/L \quad \Longrightarrow \quad \text{bound} = \frac{2L\Delta_0 B}{N} + \frac{\sigma^2}{B}.$$

At A Plateau: Step Size Or Batch Size?

Smooth strongly convex setting, independent batch size B, $0 < \eta \leq 1/L$:

$$\mathbb{E}[\Delta_T] \leq (1 - \mu\eta)^T \Delta_0 + \frac{L\eta\sigma^2}{2\mu B}.$$

Change	Noise Bound	Contraction Factor	Samples / Update
$\eta \to \eta/4$	$\div 4$	$1 - \mu\eta/4$	B
$B \to 4B$	$\div 4$	$1 - \mu\eta$	$4B$

References

Proofs And Further Reading

Bubeck (2015).

Convex Optimization: Algorithms And Complexity, Sections 6.1–6.2.

Bottou, Curtis & Nocedal (2018).

Optimization Methods For Large-Scale Machine Learning.

Ghadimi & Lan (2013).

Stochastic First- And Zeroth-Order Methods For Nonconvex Stochastic Programming.

Nemirovski, Juditsky, Lan & Shapiro (2009).

Robust Stochastic Approximation Approach To Stochastic Programming.